SeungYoon Lee

seoungyoon14@gmail.com | GitHub | LinkedIn | Technical Blog

Education

Sangmyung University Seoul, Republic of Korea

B.S. Candidate, Department of Human-Centered Artificial Intelligence

Mar. 2023 – Present

Research Interests

Large Language Models (LLMs) · Retrieval-Augmented Generation (RAG) · AI Evaluation · Machine Learning · Reinforcement Learning · Healthcare AI

Research Experience

Night-to-Night Sleep Duration Variability and Daytime Sleepiness: An Actigraphy Study Using MESA Data *Undergraduate Research Project, 2026*

- Developed an end-to-end sleep data analysis pipeline using Multi-Ethnic Study of Atherosclerosis (MESA) actigraphy data, transforming 30-second epoch-level recordings into participant-level 7-day sleep variability features.
- Derived night-to-night variability measures including sleep duration variability, sleep efficiency variability, wake after sleep onset (WASO), and sleep midpoint variability.
- Validated derived sleep metrics against official MESA summary variables to assess the reliability and consistency of the preprocessing pipeline.
- Conducted multivariable regression analyses with demographic and clinical covariates including age, sex, BMI, and apnea-hypopnea index (AHI).
- Performed FDR correction, multicollinearity diagnostics, quartile-based analysis, and nonlinear sensitivity analyses to evaluate robustness.
- Found that greater night-to-night variability in sleep duration was associated with slightly higher daytime sleepiness independent of average sleep duration and major covariates.

Selected Projects

Turini — Retrieval-Augmented Generation System for Financial Education *Team Project, 2026*

- Developed a Retrieval-Augmented Generation pipeline combining dense retrieval, BM25-based sparse retrieval, Reciprocal Rank Fusion (RRF), reranking, and LLM-based response generation.
- Designed a hybrid retrieval architecture to improve evidence retrieval across financial and economic educational content.
- Constructed and refined an evaluation dataset containing 157 questions and 298 gold evidence annotations.
- Evaluated retrieval quality using Hit@K, Recall@K, MRR, and nDCG.
- Conducted controlled experiments on chunk size and overlap, selecting an optimized chunking configuration based on retrieval performance.
- Analyzed retrieval and reranking failure cases to identify cases where reranking improved or degraded evidence ranking.
- Extended the system toward multi-turn conversational RAG by incorporating dialogue context into retrieval and response generation.

LLM / RAG Evaluation Framework *Personal Project, 2026*

- Developed an evaluation framework for detecting performance regressions in LLM- and RAG-based applications.
- Implemented classification and retrieval evaluation using Accuracy, Macro F1, Hit@K, Recall@K, MRR, and nDCG.
- Designed baseline-versus-candidate workflows to identify regressions caused by model, prompt, retrieval, or reranker changes.
- Integrated automated regression testing with GitHub Actions CI and evaluated LLM intent classification using BANKING77.

Reinforcement Learning-Based Penalty Kick Agent *Unity ML-Agents Project, 2025*

- Developed a reinforcement learning environment in Unity for training an autonomous agent to perform penalty kicks.
- Designed the agent's observation space, action space, reward function, and training environment.

- Applied reward shaping and iterative environment tuning to improve learning stability and goal-oriented behavior.
- Trained and evaluated policies using Unity ML-Agents and analyzed agent behavior across training iterations.

Competitions & Activities

EXAONE Model Quantization Hackathon(2026) *LG*

- Participated in a hackathon focused on optimizing the EXAONE large language model through aggressive model quantization.
- Explored sub-2-bit quantization approaches to reduce memory and computational requirements while maintaining model performance.
- Investigated trade-offs between model compression, inference efficiency, and downstream performance.

Quantum Information Hackathon(2023) *Quantum Information Research Support Center (QCenter)*

- Participated in a quantum computing hackathon organized by the Quantum Information Research Support Center (QCenter).
- Explored a Quantum Recurrent Neural Network (QRNN) approach for sequential learning tasks.

Technical Skills

Programming Languages: Python, C, C#, SQL, TypeScript

Machine Learning / AI: scikit-learn, XGBoost, LightGBM, Reinforcement Learning, Unity ML-Agents, LLM Evaluation, Model Quantization

LLM / Retrieval Systems: Retrieval-Augmented Generation, Dense Retrieval, BM25, Hybrid Retrieval, Reciprocal Rank Fusion, Reranking, FAISS, LLM-as-a-Judge

Evaluation & Experimentation: Accuracy, Macro F1, Hit@K, Recall@K, MRR, nDCG, Regression Testing, RAG Evaluation

Data Analysis: Pandas, NumPy, Statistical Regression, Multicollinearity Diagnostics, Multiple-Comparison Correction

Development Tools: Git, GitHub, GitHub Actions, Firebase, React Native, Unity

Research & Technical Interests

- Evaluation and reliability of large language model applications
- Retrieval and reranking methods for Retrieval-Augmented Generation
- Automated regression testing for LLM and RAG systems
- Efficient language models and model compression
- Machine learning for healthcare and digital health data